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Require [u]intptr_t

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Changelog

- R3:
 - SG22 / WG14 saw this, and have no compatibility concerns with C.
 - Updated Wording to account for potential future adoption of P3348 (Update C++ to C23).
- R2:
 - [SG1 Poll]:
There are no SG1 concerns with P3248R1

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- **R1:**
 - Add "Header file inconsistency between C and C++" discussion to "Design" section.
 - Add context of C programming language efforts to require `[u]intptr_t`.
 - Add clarifications with respect to Memory Tagging.
 - Add C23 specification of `[u]intptr_t`.
 - Add impact analysis on conforming and non-conforming implementations.
- **R0:** initial draft.

Motivation

Proposals like P2835 (<https://wg21.link/p2835>) and P3125 (<https://wg21.link/P3125>) use `[u]intptr_t` as an integer type capable of holding a pointer value in their APIs^[1]. However, `[u]intptr_t` being *optional* forces sub-optimal design choices such as making APIs optional or introducing workarounds.

The potential absence of `[u]intptr_t` compromises the portability of high-level software and attempts to address this introduce software engineering overheads and potential portability bugs, as seen in libvlc PR#1519 (https://code.videolan.org/videolan/vlc/-/merge_requests/1519).

This proposal advocates for requiring `[u]intptr_t` in C++ to ensure that all C++ code can rely on integer types capable of holding a pointer value.

Status quo

C Programming language semantics of `[u]intptr_t`

The ISO/IEC 9899:2023 Working Draft (<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3096.pdf>) specifies `[u]intptr_t` semantics as follows:

7.22.1.4 Integer types capable of holding object pointers

1. The following type designates a signed integer type, other than a bit-precise integer type, with the property that any valid pointer to `void` can be converted to this type, then converted back to pointer to `void`, and the result will compare equal to the original pointer

`intptr_t`

The following type designates an unsigned integer type, other than a bit-precise integer type, with the property that any valid pointer to `void` can be converted to this type, then converted back to pointer to `void`, and the result will compare equal to the original pointer:

`uintptr_t`

These types are optional.

Other sections of the specification provide additional operations that preserve `[u]intptr_t` values:

- `memcpy`
- I/O functions like `fprintf / fscanf` on `[u]intptr_t`.

ISO/IEC CD TS 6010 - A provenance-aware memory object model for C

(<https://www.iso.org/standard/81899.html>) (N3005 (<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3005.pdf>)) explores extending these guarantees.

C++'s `[expr.reinterpret.cast#5]` (<https://eel.is/c++draft/expr.reinterpret.cast#5>) brings C `[u]intptr_t` semantics into C++ as follows:

A value of integral type or enumeration type can be explicitly converted to a pointer. A pointer converted to an integer of sufficient size (if any such exists on the implementation) and back to the same pointer type will have its original value ([basic.compound]); mappings between pointers and integers are otherwise implementation-defined.

C++ `[cstdio.syn#1]` (<https://eel.is/c++draft/cstdio.syn#1>) imports `fprintf / fscanf` from C.

Requiring `[u]intptr_t` in the C Programming Language

The C programming language proposal N2889 (<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n2889.htm>) explored requiring `[u]intptr_t`. It was rejected for C23 but adopted into ISO/IEC CD TS 6010 - A provenance-aware memory object model for C

(<https://www.iso.org/standard/81899.html>) (N3005 (<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n3005.pdf>)) to enable C to gain experience with the proposal. There is consensus that this the right approach, but there is not enough implementation experience.

Impact analysis

Impact on conforming implementations

A survey found ubiquitous support for `[u]intptr_t` in *conforming* C++ implementations (*):

- C++ Standard Library implementations assume `[u]intptr_t` are available: libstdc++, libc++, and Microsoft STL.
- C++ Compilers supporting `[u]intptr_t` on all targets, including those with non-standard pointer sizes: GCC, Clang, MSVC.
- C++ Platform ABIs specify the size and alignment of pointers and the calling convention of Integer types, fixing the ABI of `[u]intptr_t`. *Extended integer types* avoid breaking the ABI of `intmax_t` when introducing a wider `[u]intptr_t` (this used to be a problem, see N2889 (<https://www.open-std.org/jtc1/sc22/wg14/www/docs/n2889.htm>)).

We did not find any *conforming* implementation that is inconsistent in C and C++ with respect to the availability of `[u]intptr_t`: all implementations found provide these types in the headers of both programming languages.

We did not find any *conforming* implementation that:

- would stop conforming if C++ were to require `[u]intptr_t`, or
- does not already provide `[u]intptr_t`.

Therefore, we conclude that C++ requiring `[u]intptr_t`:

- does not regress current implementation support, and
- does not require any implementation effort,

for any currently conforming implementation.

(*) many C++ implementations are not conforming in one way or another, but here we focus on pointers.

Impact on non-conforming implementations

Full support for `[u]intptr_t` cannot be expected on platforms that lack full support for pointers. All the non-conforming implementations found, are non-conforming with respect to pointer support. For example, their I/O functions (`fprintf` / `fscanf`) or `memcpy` to unaligned

addresses do not uphold pointer round-trips (e.g. via `%p`) validity requirements.

We evaluate the impact on these implementations in terms of what "partial" support for `[u]intptr_t` can be provided and at what effort (e.g. at least to document which partial support, if any, is provided).

We found that the following non-conforming platforms would *not* be impacted by C++ requiring `[u]intptr_t`:

- **CHERI C++:** already provides `[u]intptr_t` documenting limitations on its support. For more details, see, e.g., the CHERI C/C++ Programming Guide (<https://www.cl.cam.ac.uk/techreports/UCAM-CL-TR-947.pdf>) or the more recent: *Zaliva et al.*, Formal Mechanised Semantics of CHERI C: Capabilities, Undefined Behaviour, and Provenance (<https://www.cl.cam.ac.uk/~pes20/asplos24spring-paper110.pdf>), ASPLOS '24.

We found that the following non-conforming platforms may be impacted by C++ requiring `[u]intptr_t`:

- **IBM i** (https://www.ibm.com/products/ibm-i?utm_content=SRCWW&p1=Search&p4=43700074687253318&p5=e&p9=58700008221000440&gclid=EAlaIQobChMI9p6L2tXBhgMVKhOtBh0RBAPyEAAyASAAEgKFSvD_BwE&gclidsrc=aw.ds) (see also IBM AS/400 (https://en.wikipedia.org/wiki/IBM_AS/400)): uses PowerPC AS Tagged Memory Extensions (<https://www.devever.net/~hl/ppcas>). Its ILE C++ compiler already documents standards compliance limitations, including lack `[u]intptr_t` (even though these types are currently optional). Whether it can implement `[u]intptr_t` is to be determined, but if it can, whether it does so may depend on other factors like customer demand.
- **Elbrus** (<https://en.wikipedia.org/wiki/MCST>) has memory tagging (https://en.wikipedia.org/wiki/Tagged_architecture) capabilities (<https://news.ycombinator.com/item?id=27494357>): in protected mode, pointers are 128-bit wide and include a memory address, an object size, and an offset, but `[u]intptr_t` is only 64-bit wide and does not support `ptr2int2ptr` round-trips (https://github.com/ivmai/libatomic_ops/issues/61). CMake (https://cmake.org/cmake/help/latest/variable/CMAKE_LANG_COMPILER_ID.html) and libfmt (<https://github.com/fmtlib/fmt/pull/3057>) support its compiler, and the latter employs a fallback (<https://github.com/fmtlib/fmt/blob/8ee6c9401471b402e2e33f4671edcf74af33e20c/include/fmt/format.h#L429>) in case `[u]intptr_t` is not available. Whether it can implement `[u]intptr_t` is to be determined, and if it can, whether it does so may depend on other factors.

Header file inconsistency between C and C++

On a platform in which the C implementation does not provide this type (only non-conforming implementations found), the `<stdint.h>` header does not provide this type in the C programming language (e.g. when processed by a C compiler).

Per [support.c.header.other.1](), in C++ `<stdint.h>` has the same content as `<cstdint>`. In a platform in which `[u]intptr_t` is not available to C via `<stdint.h>`, it is required to be available to C++ via both `<stdint.h>` and `<cstdint>`:

```
// foo.c - C compiler
#include <stdint.h> // C header processed by C compiler
intptr_t val;      // ill-formed

// bar.cpp - C++ compiler
#include <stdint.h> // C header processed by C++ compiler
intptr_t val;      // well-formed

// baz.cpp - C++ compiler
#include <cstdint>  // C++ header processed by C++ compiler
std::intptr_t val; // well-formed
```

This disconnect in platforms in which the C implementation does not provide `[u]intptr_t` may impact developer productivity in those platforms.

Design

Design alternatives:

1. C++ requires `[u]intptr_t`.
2. C++ adds new integer types - different from `[u]intptr_t` - capable of holding a pointer value.
3. Do nothing.

This proposal advocates for Option 1, i.e., for C++ to require `[u]intptr_t`, because:

- **Pre-existing code:** All implementations surveyed provide these on all platforms. This has led to a large corpus of pre-existing code using `[u]intptr_t`. Requiring `[u]intptr_t` makes this code portable to all platforms C++ supports. Inventing new C++ types would make this code non-idiomatic and cause significant churn on all ecosystems for little added value.
- **Compatibility with C:** By requiring these types in C++ with the same semantics as C we ensure C++ remains forward compatible with C eventually requiring these types, since if that were to happen, C++ would get their exact same semantics and ABI. This is particularly important with respect to the pointer provenance rules as specified in TS 6010 (<https://www.iso.org/standard/81899.html>). Adding new C++ types that are not available in C would reduce C++'s compatibility with C. There is however a nuanced header file inconsistency between C and C++ that is covered in the next section.

- **ABI:** Platforms whose ABI specifies `intmax_t` to be smaller than the platform's pointer size are allowed to provide wider `[u]intptr_t` integer types since C23 and C++23 due to extended integer type support.
- **Cost:** There is a cost to doing nothing. Significant time was spent on `atomic_ref::address` to find a sub-optimal solution when the right solution everyone agrees on is `uintptr_t`.

Usage Guideline

`[u]intptr_t` is well suited for C++ language or C++ Standard Library APIs that need an integer type capable of holding a *pointer value*, i.e., an integer type with a lossless conversion from/to pointer.

Some features or APIs may only need an integer type capable of holding a *pointer address*. C and C++ do not currently provide an integer type suited for this use case, but some implementations do provide it as an extension, in platforms where this distinction is crucial, e.g., CHERI C/C++ implementations provide `ptraddr_t` in `<stddef.h>` (the CHERI C/C++ Programming Guide (<https://www.cl.cam.ac.uk/techreports/UCAM-CL-TR-947.pdf>) is currently outdated and mentions `vaddr_t` instead of `ptraddr_t`).

Wording changes

Modify `[cstdint.syn]` (<https://eel.is/c++draft/cstdint.syn#1>):

1. The header `<cstdint>` supplies integer types having specified widths, and macros that specify limits of integer types.

```

// all freestanding
namespace std {
    using int8_t      = signed integer type;    // optional
    using int16_t     = signed integer type;    // optional
    using int32_t     = signed integer type;    // optional
    using int64_t     = signed integer type;    // optional
    using intN_t      = see below;              // optional

    using int_fast8_t = signed integer type;
    using int_fast16_t = signed integer type;
    using int_fast32_t = signed integer type;
    using int_fast64_t = signed integer type;
    using int_fastN_t = see below;              // optional

    using int_least8_t = signed integer type;
    using int_least16_t = signed integer type;
    using int_least32_t = signed integer type;
    using int_least64_t = signed integer type;
    using int_leastN_t = see below;             // optional

    using intmax_t     = signed integer type;
    using intptr_t     = signed integer type;   // optional

    using uint8_t      = unsigned integer type; // optional
    using uint16_t     = unsigned integer type; // optional
    using uint32_t     = unsigned integer type; // optional
    using uint64_t     = unsigned integer type; // optional
    using uintN_t      = see below;            // optional

    using uint_fast8_t = unsigned integer type;
    using uint_fast16_t = unsigned integer type;
    using uint_fast32_t = unsigned integer type;
    using uint_fast64_t = unsigned integer type;
    using uint_fastN_t = see below;            // optional

    using uint_least8_t = unsigned integer type;
    using uint_least16_t = unsigned integer type;
    using uint_least32_t = unsigned integer type;
    using uint_least64_t = unsigned integer type;
    using uint_leastN_t = see below;           // optional

    using uintmax_t    = unsigned integer type;
    using uintptr_t    = unsigned integer type; // optional
}

#define INTN_MIN      see below
#define INTN_MAX      see below
#define UINTN_MAX     see below

```



```

#define INT_FASTN_MIN      see below
#define INT_FASTN_MAX      see below
#define UINT_FASTN_MAX     see below

#define INT_LEASTN_MIN     see below
#define INT_LEASTN_MAX     see below
#define UINT_LEASTN_MAX    see below

#define INTMAX_MIN         see below
#define INTMAX_MAX         see below
#define UINTMAX_MAX        see below

#define INTPTR_MIN         see below          // optional
#define INTPTR_MAX         see below          // optional
#define UINTPTR_MAX        see below          // optional

#define PTRDIFF_MIN        see below
#define PTRDIFF_MAX        see below
#define SIZE_MAX           see below

#define SIG_ATOMIC_MIN     see below
#define SIG_ATOMIC_MAX     see below

#define WCHAR_MIN          see below
#define WCHAR_MAX          see below

#define WINT_MIN           see below
#define WINT_MAX           see below

#define INTN_C(value)      see below
#define UINTN_C(value)     see below
#define INTMAX_C(value)    see below
#define UINTMAX_C(value)   see below

```

2. The header defines all types and macros the same as the C standard library header `<stdint.h>` except that the types `intptr_t` and `uintptr_t` and the macros `INTPTR_MIN`, `INTPTR_MAX`, and `UINTPTR_MAX` are always defined and are not optional. See also: ISO/IEC 9899:2018, 7.20.
3. All types that use the placeholder N are optional when N is not 8, 16, 32, or 64. The exact-width types `intN_t` and `uintN_t` for N = 8, 16, 32, and 64 are also optional; however, if an implementation defines integer types with the corresponding width and no padding bits, it defines the corresponding typedef-names. Each of the macros listed in this subclause is defined if and only if the implementation defines the corresponding typedef-

name.

[Note 1: The macros INTN_C and UINTN_C correspond to the typedef-names `int_leastN_t` and `uint_leastN_t`, respectively. — end note]

If P3348 (<http://wg21.link/p3348>) is adopted (updates C++ to C23), also modify the following in `<cstdint>` :

```
#define INTPTR_WIDTH see below // optional  
#define UINTPTR_WIDTH see below // optional
```

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1. This does not imply that these proposals make correct use of these types; the Usage Guideline (<https://hackmd.io/U-X9IVCjSEqjXDcw9AjlnQ#Usage-Guideline>) section covers that. ↩